浙江大學 ZHEJIANG UNIVERSITY

ZJU–UIUC Institute

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Planning and Games

Constraint Satisfaction and Adversarial Search

Assignment 2 / Group 6

Course	AI Theory and Application
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Section I Constraint Satisfaction

Exact-cover formulation

The tiling problem is represented as exact cover. Each available board cell and each tile identity contributes one constraint. A legal placement covers the cells it occupies and the identity of its tile. Selecting placements that satisfy every constraint exactly once therefore covers the board without overlap or omission and uses each tile once.

Let C be the available cells and I the tile identities, giving the disjoint constraint set $U = C \sqcup I$. A legal placement p covers a subset $R_p \subseteq U$, and the binary variable x_p indicates whether it is selected. The tiling conditions become

$$\sum_{p:\, u \in R_p} x_p = 1 \quad (\forall u \in U), \qquad x_p \in \{0, 1\}. \tag{1}$$

Cell constraints enforce coverage without overlap; identity constraints enforce single use. Identical shapes still need separate identities. For example, sixty cells and twelve pentominoes give seventy-two constraints; each placement covers six: five cells and one identity.

Algorithm X and minimum remaining values

Exact cover can be solved by Algorithm X, a depth-first backtracking procedure.[1]

We first generate distinct rotations and reflections and enumerate placements that remain inside the board and avoid holes. Search uses the minimum remaining values heuristic: choose the uncovered constraint with the fewest available placements. After selecting a placement, remove all conflicting candidates; backtrack as soon as any constraint has no candidate. Integer bitsets reduce the cost of these set operations. Every candidate for the chosen constraint is considered, so the search is complete, although its worst-case complexity remains exponential.

If A denotes the available placements and U' the uncovered constraints, the minimum remaining values rule selects

$$c^* = \arg\min_{c \in U'} \big|\{p \in A : c \in R_p\}\big|. \tag{2}$$

After choosing a placement p covering c^*, remove the constraints in R_p and all other placements that intersect R_p. Zero candidates means failure; no uncovered constraints means success. Choosing the tightest constraint exposes failure early and allows search to switch adaptively between placing a tile and covering a cell.

Selected placements remain pairwise compatible, while uncovered constraints must remain coverable by subsequent candidates. Removing conflicts performs forward checking; backtracking restores the previous candidate state. The heuristic changes exploration order, not the solution set. Bitsets reduce per-node work without changing worst-case complexity.

[1] D. E. Knuth, *Dancing Links*, 2000. https://arxiv.org/abs/cs/0011047. Our candidate sets use integer bitsets, not the paper's doubly linked list implementation.

Traditional backtracking and forward-checking controls

Two fixed-order methods separate early dead-end detection from dynamic constraint selection. All three methods share the deduplicated placements, tile identities, conflict checks, and candidate order.

Backtracking (BT). Choose the first uncovered cell in row-major order and try each compatible placement covering it. Placements cannot overlap selected tiles or reuse a tile identity. Undo a failed choice and try the next candidate. BT checks current compatibility but does not proactively inspect every other cell or unused tile for an empty domain.

Backtracking with forward checking (FC). Keep the same first-empty-cell rule, but before expanding candidates inspect all unmet constraints. If any uncovered cell or unused tile has no available placement, backtrack immediately. This adds dead-end detection without MRV.

Algorithm X with MRV (AX). Choose the cell or tile identity with the fewest available placements. This detects empty domains and prioritizes tightly constrained choices. Fixed-order backtracking can itself be viewed as exact-cover search with a fixed column rule. The comparison distinguishes these selection policies; Algorithm X and backtracking are not unrelated algorithm families.

All three enumerate every compatible candidate for their selected constraint and are complete without resource limits.

Counting and comparison protocol

Count one node per recursive call, including the empty root, dead-end states, and successful leaf. Placement generation, filtering known conflicts, and internal heuristic checks do not add nodes. Stop at the first valid solution. The count therefore measures explored partial states, not complete tilings or candidate placements. Different methods may return different valid tilings.

Each instance and method receives the same cap of 10^6 recursive nodes. Reaching it is recorded separately, neither as unsatisfiability nor as the exact number needed to find a solution. The default solver has no such cap; it is a resource control for this experiment. Only search nodes are compared, not time. FC and MRV do different work within each node, so node reductions are not equivalent to proportional runtime reductions.

Results

Four boards were tested with dominoes, triominoes, and pentominoes. All twelve AX solutions passed the supplied correctness checker. BT and FC solutions were validated only when completed within the cap.

Table 1: Search nodes for three policies sharing the same candidates.

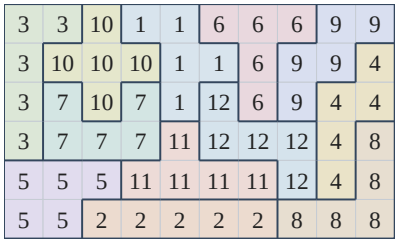
Board	Size	Candidates	BT nodes	FC nodes	AX nodes
6×10	2	3,120	31	31	31
6×10	3	3,600	85,413	69	40
6×10	5	2,056	139,426	17,902	863
5×12	2	3,090	31	31	31
5×12	3	3,520	$\geq 10^6$*	27	27
5×12	5	1,936	651,856	91,603	1,073
3×20	2	2,910	31	31	31
3×20	3	3,040	$\geq 10^6$*	31	31
3×20	5	1,236	$\geq 10^6$*	$\geq 10^6$*	238
8×8 (central hole)	2	3,000	31	31	31
8×8 (central hole)	3	3,280	40	40	28
8×8 (central hole)	5	1,568	20,224	6,150	2,104

* No solution found after 10^6 nodes; the required count remains unknown.

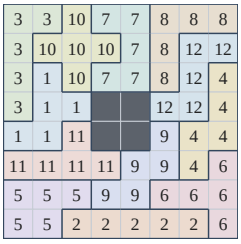
Table 2: Completion within the same node budget, across twelve instances.

Method	Solved	Node cap reached
Backtracking (BT)	9	3
Forward checking (FC)	11	1
Algorithm X + MRV (AX)	12	0

For 6×10 triominoes, BT, FC, and AX visit 85,413, 69, and 40 nodes: early dead-end detection gives the main reduction, with MRV reducing it further. For 5×12 pentominoes the counts are 651,856, 91,603, and 1,073, showing a larger effect from constraint selection. On 3×20 pentominoes, BT and FC hit the cap while AX solves in 238 nodes. All domino instances take 31 nodes for every method, with no backtracking for a heuristic to avoid. These results use a fixed candidate order; MRV need not reduce nodes on every possible tiling problem.



(a) 6×10 board



(b) Board with a central hole

Figure 1: Pentomino tilings. Equal labels identify cells belonging to one tile.

Section II Predefined Agents

Rules and search settings

The game ends when either player first completes a line on any local board. Local boards are indexed from 0 to 8 in row-major order. A move's position within its local board determines the opponent's next local board. If that board is full, the opponent may choose any empty global cell. All four games start on board 4. Offensive X moves first in games 1–2; defensive O moves first in games 3–4.

The predefined evaluation has three ordered tiers. Terminal states receive 10000 for an X win and -10000 for an O win. Otherwise, X scores 500 per unblocked two-in-a-row and 100 per blocked O two-in-a-row. O scores -100 per unblocked two-in-a-row and -500 per blocked X two-in-a-row. Only when the second tier has no counts are occupied local corners scored: 30 each for X and -30 each for O. Rows, columns, and diagonals are counted separately, so a cell may contribute to more than one line.

Both algorithms search three plies and stop early at terminal states. Leaves use the root agent's evaluation throughout a search, always expressed as X utility: X maximizes and O minimizes. Moves are visited in row-major order, and the current best move is replaced only by a strict improvement. Thus ties retain the first move. Alpha–beta search prunes when $\alpha \geq \beta$.

Minimax recurrence and pruning

A state must include the global board B, the forced local board k, and the player p: $s = (B, k, p)$. Piece locations alone do not determine legal actions. A move at global coordinates (r, c) sets $k' = 3(r \bmod 3) + (c \bmod 3)$.

Let $\mathcal{A}(s)$ be the legal actions, $T(s, a)$ the successor, and d the remaining depth. For a fixed root evaluation E,

$$V_d(s) = \begin{cases} u(s), & s \text{ is terminal}, \\ E(s), & d = 0 \text{ and } s \text{ is nonterminal}, \\ \max_{a \in \mathcal{A}(s)} V_{d-1}(T(s,a)), & p = X, \\ \min_{a \in \mathcal{A}(s)} V_{d-1}(T(s,a)), & p = O. \end{cases} \tag{3}$$

Here $u(s)$ is -10000, zero, or 10000 for an X loss, draw, or win. Minimax assumes the opponent chooses the action most adverse to this utility. The resulting conservative decision is optimal for the truncated tree, not necessarily for the full game, since nonterminal E is approximate.

Alpha and beta record existing MAX guarantees and MIN restrictions along the path. Suppose MAX already has an option worth 5. If another MIN branch finds a child worth 3, that branch's final value cannot exceed 3, so its remaining children cannot improve MAX's choice. Pruning preserves the minimax value for the same depth and evaluation.[2] For uniform branching factor b and depth d, ordinary search costs $O(b^d)$. With ideal move ordering, pruning can approach $O(b^{d/2})$; its worst case remains $O(b^d)$. The required row-major order does not justify assuming ideal ordering in these experiments.

[2] D. E. Knuth and R. W. Moore, *An Analysis of Alpha-Beta Pruning*, Artificial Intelligence 6 (1975), 293–326. https://doi.org/10.1016/0004-3702(75)90019-3.

Outcomes and pruning

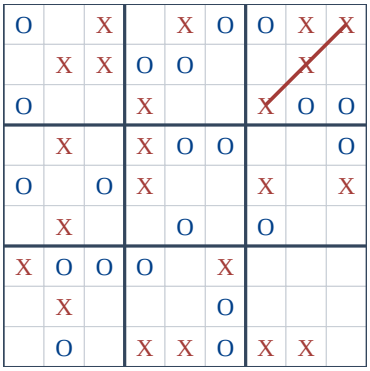
Node counts include visited roots and leaves and are summed over all turns. In Table 3, MM denotes minimax and AB denotes alpha–beta.

Table 3: Results of the four required matchups.

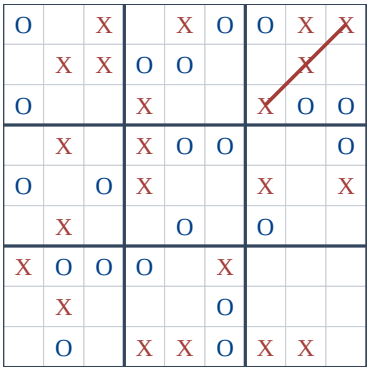
Game	X	O	First	Plies	X nodes	O nodes	Total	Win
1	MM	MM	X	43	8768	7948	16716	X
2	MM	AB	X	43	8768	3398	12166	X
3	AB	MM	O	44	3353	8528	11881	X
4	AB	AB	O	44	3353	3431	6784	X

X won all four games. Games 1–2 have identical move sequences, with a 27.2% reduction in visited nodes after pruning. Games 3–4 also have identical sequences, with a 42.9% reduction. Alpha–beta therefore preserved minimax decisions while reducing search effort in these matchups. Games 1 and 4 cannot isolate the effect of pruning because they have different starting players.

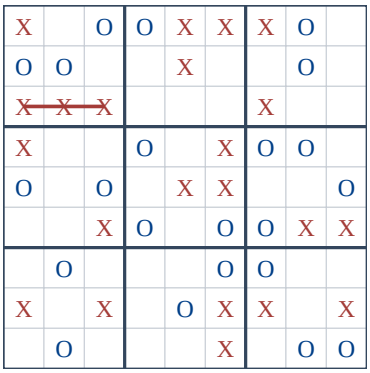
The offensive heuristic emphasizes creating its own threats, whereas the defensive heuristic emphasizes completed blocks. The latter does not directly measure the danger of the board to which the next player is sent, which can leave it vulnerable to sequences of forced routing.



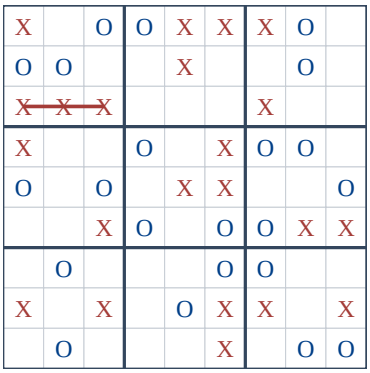
(a) Game 1: X wins; 16,716 nodes



(b) Game 2: X wins; 12,166 nodes



(c) Game 3: X wins; 11,881 nodes



(d) Game 4: X wins; 6,784 nodes

Figure 2: Final boards of the predefined agents.

X is offensive and O is defensive. Thick lines separate local boards; colored lines mark winning triples. The four boards correspond to the rows of Table 3.

Section III Designed Evaluation

Formulation

The designed agent plays O against predefined offensive X. Both use depth-three alpha–beta search. The evaluation considers both players' threats and the board where the next player must actually move. A winning opportunity elsewhere is not necessarily playable on the next turn, so routing is evaluated explicitly.

Let T count unblocked two-in-a-row lines, W sum distinct winning empty cells over local boards, and F count local boards with at least two distinct winning cells. Let C count occupied corners and M occupied centers, with subscripts X and O identifying the player. Set $s = 1$ when X moves next and $s = -1$ when O moves next. Of the next player's L legal moves, S are safe: they do not immediately send the opponent to a board where that opponent can win. When a move routes back to the same local board, its cell is updated before testing the danger.

$$H = 400(T_X - T_O) + 35(W_X - W_O) + 800(F_X - F_O) \\ + 15(C_X - C_O) + 40(M_X - M_O) + 6000s\frac{S}{L}.$$

Actual terminal states take priority and return ± 10000. If the next player can win immediately, return $9500s$. If every legal move allows an immediate opponent win, return $-9000s$. Otherwise, use H clipped to $[-8000, 8000]$. A full global board with no winner returns zero. These static tactical checks do not extend the recursive search beyond three plies.

The ratio S/L avoids favoring a board merely because it has fewer empty cells. The feature F distinguishes two independent winning cells from two lines sharing one blocking cell. Immediate threats take priority over general positional scores; centers and corners have smaller weights.

Feature interpretation and the search horizon

T measures potential attacking lines; W and F distinguish threats that can be removed by a single block from independent winning cells. Two lines sharing one winning square can still be blocked together. Even independent threats depend on the forced board, so global line counts alone are insufficient.

The safety term has the current player's sign. A larger S/L increases X utility when X moves next and decreases it when O moves next. It converts the consequence of sending the opponent to a particular board into a comparable state feature. Separating actual outcomes, immediate tactical values, and ordinary heuristic values prevents distant line or corner scores from outweighing immediate wins and losses. These scores rank positions; they are not winning probabilities.

Finite depth creates a horizon effect: longer forced returns can fall beyond the search boundary. The evaluation recognizes immediate wins and positions where every choice loses immediately, but it cannot prove the safety of longer sequences. Match results are therefore necessary, and success against the fixed opponent does not establish strength against arbitrary human strategies.

Parameter comparison

Development covered all nine starting boards and both starting players, giving eighteen conditions. The initial safe-count scheme won eleven; the best adjusted safe-count scheme won fourteen. Position weights, threat spread, and safety ratios were then compared. The ratio weight 6000 was selected for the most development wins in Table 4. The features and candidate weights were chosen manually, followed by a limited parameter sweep. The winning-cell weight 35 and position weights 15 and 40 were retained as heuristic settings, not established as optimal. The extended-rule weights were also chosen manually and held fixed across its depth comparisons.

Table 4: Development comparison of mobility features.

Mobility feature	Weight	Development wins / 18
Safe-move count	35	11
Safe-move count	1000	14
Safe-move ratio	1000	16
Safe-move ratio	3000	16
Safe-move ratio	6000	18

There are only eighteen opening conditions, so development and subsequent random games are not strictly independent. The results establish an improvement against this specified opponent, rather than a guarantee against arbitrary strategies.

Randomized match results

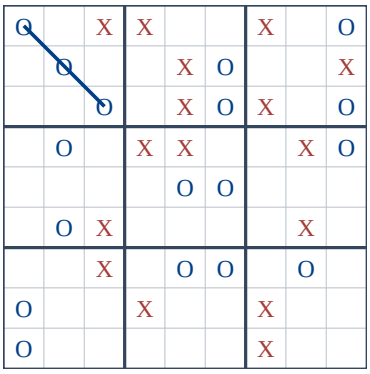
With random seed 448, each game's starting board and player were sampled randomly. Across 40 games, O won 40, X won 0, and 0 were drawn. The designed agent's win rate was 100.0%, exceeding the required 80%. The average total search count was 6,516.7 nodes per game.

Policies and tie-breaking are deterministic. Repeated starting conditions therefore produce repeated games; the forty runs are not forty distinct tests of opponent strategy.

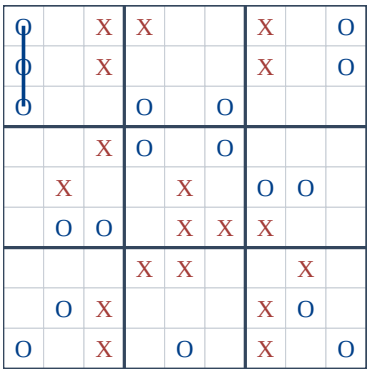
Table 5: Per-game results; node counts sum both players' searches.

Game	Start	First	Plies	Nodes	Win
1	4	O	33	5,759	O
2	1	X	36	7,901	O
3	4	O	33	5,759	O
4	0	X	32	6,700	O
5	8	O	27	5,217	O
6	6	X	36	7,107	O
7	1	X	36	7,901	O
8	3	X	38	6,716	O
9	8	O	27	5,217	O
10	5	X	36	6,536	O
11	5	O	41	6,969	O
12	7	X	36	6,206	O
13	4	X	34	6,104	O
14	8	O	27	5,217	O
15	5	O	41	6,969	O
16	5	X	36	6,536	O
17	1	X	36	7,901	O
18	0	X	32	6,700	O
19	0	O	35	7,409	O
20	5	X	36	6,536	O
21	3	X	38	6,716	O
22	8	O	27	5,217	O
23	3	O	29	5,976	O
24	8	O	27	5,217	O
25	4	X	34	6,104	O
26	6	X	36	7,107	O
27	6	O	37	7,187	O
28	7	X	36	6,206	O
29	6	O	37	7,187	O
30	1	O	33	6,850	O
31	6	O	37	7,187	O
32	5	X	36	6,536	O
33	3	O	29	5,976	O
34	0	X	32	6,700	O
35	6	X	36	7,107	O
36	3	X	38	6,716	O
37	0	O	35	7,409	O
38	8	O	27	5,217	O
39	3	O	29	5,976	O
40	3	X	38	6,716	O

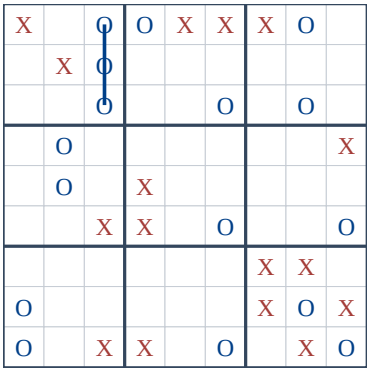
Representative positions



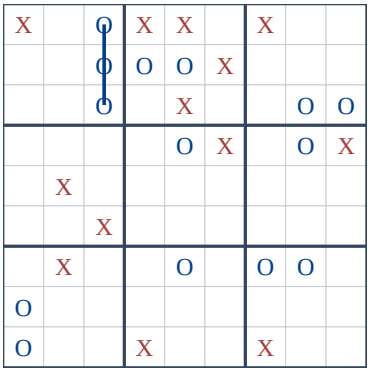
(a) Game 1: start 4, O first



(b) Game 2: start 1, X first



(c) Game 4: start 0, X first



(d) Game 5: start 8, O first

Figure 3: Representative final boards of the designed agent.

Immediately before the final move, X had 5, 7, 3, 4 legal choices in the four games, respectively, but none could prevent O from winning on the next turn. The preceding threats and routing had already forced the outcome; these wins did not merely depend on X overlooking a final block. The predefined offensive agent emphasizes its own two-in-a-row score, whereas the designed evaluation also considers the destination of the next turn and gives routing threats greater priority. Final boards alone cannot isolate the causal effect of an individual weight. These findings remain specific to the tested opponent and opening conditions.

Section IV Human Games

Protocol and game records

The human plays X and the designed agent plays O, with randomized starting board and player. Twelve games were played with depth-three O. Win rates and representative final boards are used to analyze the agent's strengths and weaknesses.

12 human games are recorded.The human did not win every game in the sample.

Table 6: Human-game records. Starting boards are numbered 0–8.

Game	Start	First	Plies	Winner	Agent nodes	Status
1	4	O	31	Agent	2,209	Recorded
2	1	X	24	Agent	2,739	Recorded
3	4	O	33	Agent	3,832	Recorded
4	0	X	20	Agent	2,160	Recorded
5	8	O	15	Agent	1,723	Recorded
6	6	X	41	Human	3,825	Recorded
7	1	X	30	Agent	2,744	Recorded
8	3	X	43	Human	3,449	Recorded
9	8	O	41	Agent	4,036	Recorded
10	5	X	10	Agent	971	Recorded
11	5	O	11	Agent	1,333	Recorded
12	7	X	34	Agent	3,473	Recorded

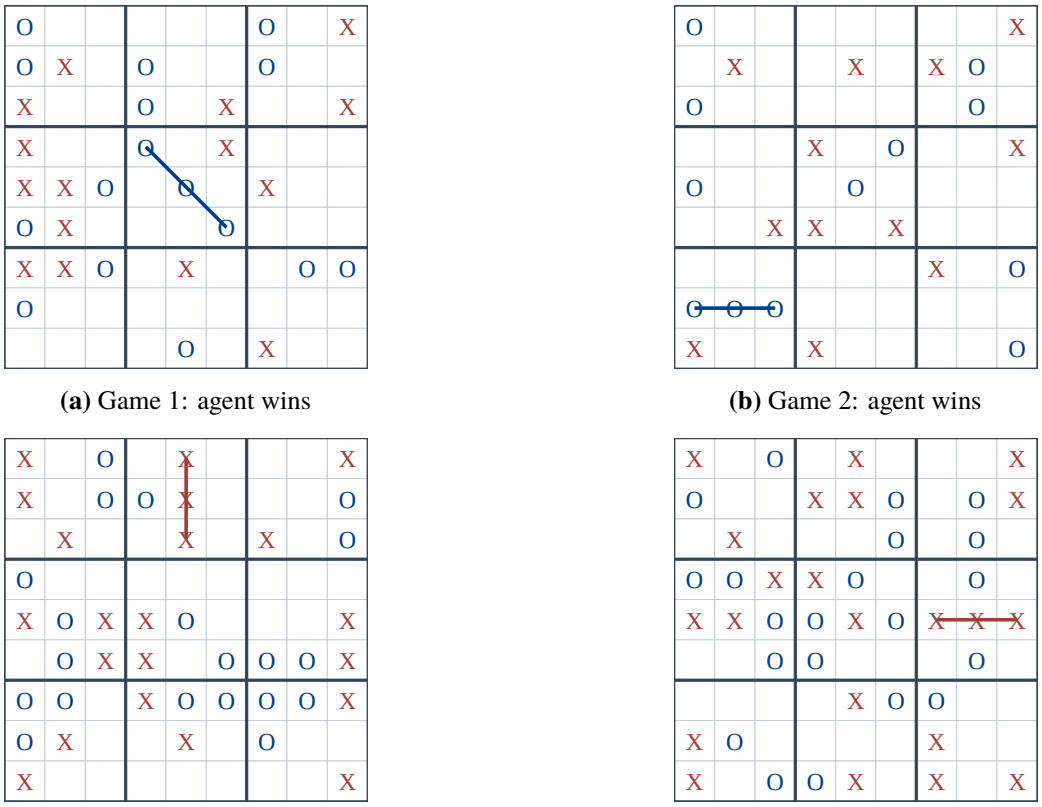
Win-rate summary

Table 7: Summary statistics for the human experiment.

Measure	Result
Completed games	12 / 10
Human wins / agent wins / draws	2 / 10 / 0
Human win rate	16.7%
Agent win rate	83.3%

Each completed game records the starting board, first player, number of plies, winner, agent node count, and final board. Human win rate is human wins divided by completed games; a draw is not a win. Unstarted or interrupted games do not enter the denominator.

Representative boards and observations



(a) Game 1: agent wins

(b) Game 2: agent wins

(c) Game 6: human wins

(d) Game 8: human wins

Figure 4: Representative human-game boards, including wins by both sides.

Outcome distribution and whether humans always won

Neither side won every game. The human won games 6 and 8; the agent won the other recorded games. Human performance differs from the synthetic opponents' zero win rate: success against fixed policies does not establish the same advantage against people.

Critical moves, strengths, and mistakes

Games 1, 2, 6, and 8 cover both outcomes. Enumerating every legal penultimate move shows that all four human choices in game 1 allowed an immediate agent win. In game 2, four of five choices avoided immediate defeat, but the played move did not. In games 6 and 8, none of the agent's five and three choices, respectively, could prevent the human's next-move win. Those defeats were already unavoidable at the penultimate move.

Evaluation weaknesses and improvements

Depth-three search and immediate safety ratios do not guarantee long-term routing safety. The losses expose weaknesses, although final boards cannot identify an individual weight's causal effect. Compare longer-route and threat features at the same depth and validate against new automatic opponents.

These games involve one human player and a small sample; they do not estimate the general population's win rate.

Supplementary synthetic opponents (not human play)

Thirty automated games compare three synthetic policies against depth-three designed O. Synthetic X uses the same first-local-win rules. Random X samples legal moves uniformly. Tactical X first takes an immediate win, then blocks an immediate opposing win in the legal area, then prefers centers and corners; with probability 15% it instead moves randomly. Shallow X uses depth-two alpha–beta with the predefined offensive evaluation, moving randomly with probability 10%. Tactical ties are random; search retains the first tie.

Seed 20261106 generates ten starting boards and first-player choices, shared across policies. Game i of policy p uses seed $20261106 + 1000p + i - 1$, both indices starting at one. Policies and error probabilities were fixed before execution. All outcomes were retained without screening losses or selecting new openings.

Table 8: Synthetic opponents versus designed O; not human statistics.

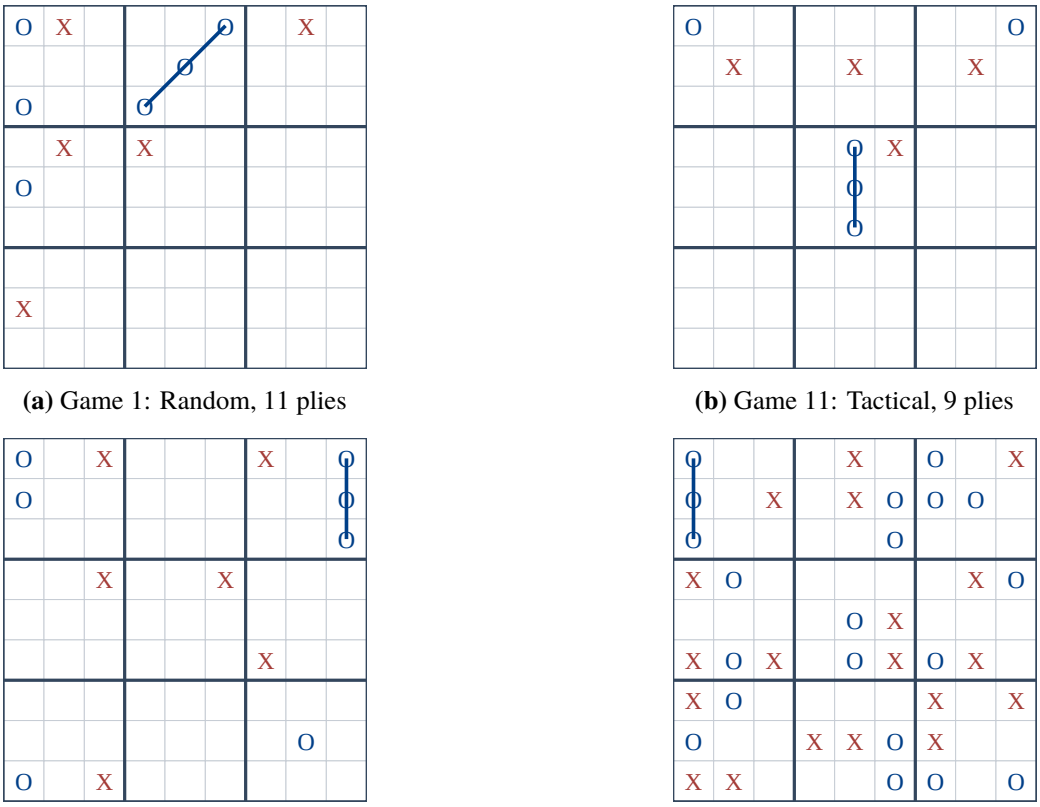
Synthetic X	Games	X wins	O wins	Draws	O win rate	Mean plies
Random	10	0	10	0	100%	15.9
Tactical	10	0	10	0	100%	9.1
Shallow	10	0	10	0	100%	29.5

Table 9: All thirty synthetic games. Starting boards are numbered 0–8.

Game	Policy	Start	First	Plies	Win
1	Random	0	O	11	O
2	Random	2	O	25	O
3	Random	0	X	28	O
4	Random	5	O	13	O
5	Random	4	X	14	O
6	Random	0	X	8	O
7	Random	0	X	10	O
8	Random	1	O	21	O
9	Random	4	X	14	O
10	Random	2	O	15	O
11	Tactical	0	O	9	O
12	Tactical	2	O	7	O
13	Tactical	0	X	10	O
14	Tactical	5	O	11	O
15	Tactical	4	X	10	O

Game	Policy	Start	First	Plies	Win
16	Tactical	0	X	8	O
17	Tactical	0	X	10	O
18	Tactical	1	O	11	O
19	Tactical	4	X	6	O
20	Tactical	2	O	9	O
21	Shallow	0	O	13	O
22	Shallow	2	O	33	O
23	Shallow	0	X	36	O
24	Shallow	5	O	37	O
25	Shallow	4	X	38	O
26	Shallow	0	X	10	O
27	Shallow	0	X	36	O
28	Shallow	1	O	17	O
29	Shallow	4	X	36	O
30	Shallow	2	O	39	O

Representative synthetic games and interpretation



(a) Game 1: Random, 11 plies

(b) Game 11: Tactical, 9 plies

(c) Game 21: Shallow, 13 plies

(d) Game 30: Shallow, 39 plies

Figure 5: Representative synthetic-opponent boards (not human play).

The boards show the first game of each policy and the longest game overall. Each synthetic policy lost all ten games: designed O won 30/30, or 100%. Mean lengths were 15.9, 9.1, and 29.5 plies, respectively. Shallow search delayed termination on these openings. Tactical X checks immediate threats in the legal area, while its positional preferences ignore routing risk. Even an immediate block can therefore send the opponent to another board with an existing threat. Local defense alone does not resolve cross-board routing; game length itself is not a ranking of playing strength.

Every trace was replayed to check legal moves, board changes, and outcomes, then reproduced with the same seed to verify the full trajectory. There are only ten games per policy, with shared openings: these thirty games are not independent human samples. Zero losses establish an advantage on this fixed test, not a guarantee against arbitrary opponents.

Extra Credit

Global-line rules

A local three-in-a-row captures and closes that board; no further move is allowed inside it. A full local board without a winner is marked D and also closed. A move routes the opponent as before, but a closed destination permits free choice among all open local boards. Victory requires three captured boards in a global row, column, or diagonal. The game is drawn when all local boards are closed without a global winner.

The state must track local ownership as well as pieces. Let $z_i \in \{X, O, D, _\}$ denote an X capture, O capture, closed draw, or open board. The capture total alone is insufficient: two disconnected captures need not be as valuable as two that create an open global threat.

Custom offensive and defensive evaluations

In the twelve games, both offensive X and defensive O use depth-three alpha–beta search. Let G count captured boards, P, Q count open global two-mark and one-mark lines, and T, L count open two-mark and one-mark lines within open local boards. K weights captured boards by 3 at the center, 2 at corners, and 1 at edges. A counts two-mark lines on local boards available to the next player. J equals that player's sign s if an immediate local capture is possible, and zero otherwise. In X utility,

$$\begin{aligned} H_g = {} & aG_X - 450G_O + bP_X - 1400P_O + 150Q_X - 120Q_O \\ & + dT_X - 35T_O + 4(L_X - L_O) + 25(K_X - K_O) \\ & + 80(A_X - A_O) + 200J. \end{aligned}$$

The offensive root uses $(a, b, d) = (500, 1600, 45)$; the defensive root uses $(550, 1800, 55)$. By weighting X captures and threats more heavily, the defensive agent prioritizes limiting X while minimizing the score. Actual global outcomes return ± 10000. An immediate global win for the next player returns $9500s$; otherwise H_g is clipped to $[-8000, 8000]$, with zero for a global draw. Closed boards contribute no local potential, and D blocks global attacking lines.

Experimental results

With random seed 2013, 12 games gave 5 X wins (41.7%), 5 O wins (41.7%), and 2 draws (16.7%). The mean game lasted 54.2 plies and visited 10,973.4 nodes.

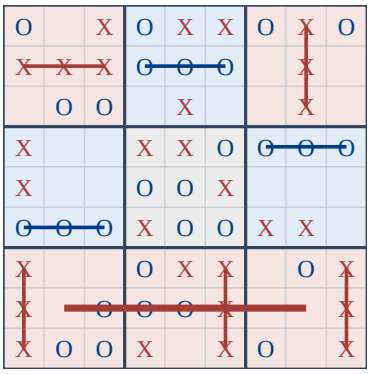
Table 10: Global-line results. Capture counts refer to won local boards at termination.

Game	Start	First	Plies	X boards	O boards	Nodes	Win
1	6	X	57	5	3	10,425	X
2	0	X	55	4	5	12,530	Draw
3	6	X	57	5	3	10,425	X
4	5	X	54	5	4	9,343	O
5	6	X	57	5	3	10,425	X
6	8	O	48	5	3	12,700	X
7	6	X	57	5	3	10,425	X
8	8	X	52	2	5	16,189	O
9	4	X	56	3	6	10,004	O
10	0	O	49	3	6	8,712	O
11	6	O	47	3	4	10,666	O
12	1	X	61	5	3	9,837	Draw

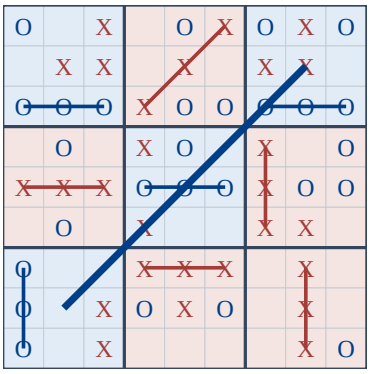
All games were replayed step by step: every move followed the forced-board rule, closed boards remained unchanged, and the recorded outcome matched the global-line test. Search used row-major order and retained the first move on ties. Both agents won and lost in the twelve games, with draws also occurring. Local success therefore did not always convert into a global win. Repeated opening conditions produced repeated traces, so these rates describe this deterministic pair of policies.

The global heuristic must balance local captures against their arrangement. Larger global-threat weights prioritize boards needed to complete a global line, but fixed depth can still miss longer routing sequences. This experiment demonstrates the implemented rules and runnable strategies; it does not establish that either weighting is superior against broader opponents.

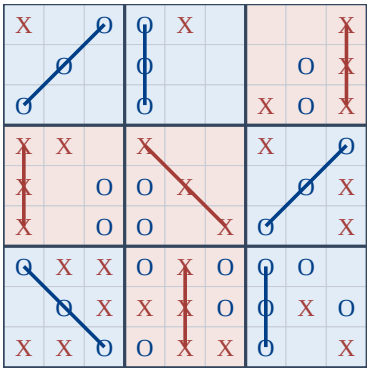
Representative extended-game boards



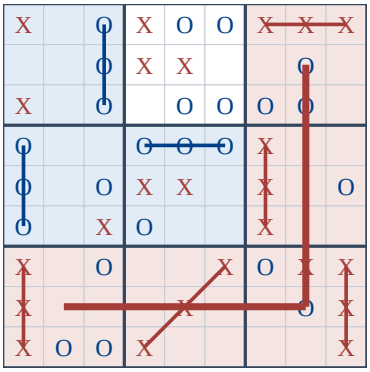
(a) Game 1: X wins; boards 5:3



(b) Game 4: O wins; boards 5:4



(c) Game 2: draw; boards 4:5



(d) Game 6: X wins; boards 5:3

Figure 6: Representative final boards under the global-line rule.

Pale red and blue indicate captured, closed X and O boards; gray marks drawn local boards and white marks open boards. A thick line across three local-board centers marks a global win.

In game 1, X won through local boards 6,7,8. The capture count was 5:3; the arrangement of captured boards, not their total alone, determines victory.

In game 4, O won through local boards 2,4,6. The capture count was 5:4; the arrangement of captured boards, not their total alone, determines victory.

Game 2 ended in a draw: every local board was closed, but neither side formed a global line. 0 local boards were drawn.

In game 6, X won through local boards 6,7,8. The capture count was 5:3; the arrangement of captured boards, not their total alone, determines victory.

Depth controls under the extended rules

Keep the custom offensive X and custom defensive O weights unchanged. Let (d_X, d_O) denote their search depths; compare $(3,3)$, $(3,4)$, $(3,5)$, $(4,4)$, and $(5,5)$. Each covers all nine starting boards and both first players, giving ninety games. Unlike the original twelve random games, every opening occurs once per configuration. Compare depths within these matched eighteen openings. X and O use the custom offensive and defensive evaluations described above, respectively.

Table 11: Global-line depth controls; eighteen games per row. Nodes sum both players.

d_X	d_O	X wins	O wins	Draws	O win rate	Mean nodes	Ref. time/s
3	3	6	6	6	33.3%	10,817	1.47
3	4	4	11	3	61.1%	25,785	3.32
3	5	3	13	2	72.2%	99,782	14.24
4	4	8	4	6	22.2%	41,147	4.86
5	5	4	5	9	27.8%	203,091	24.81

Against depth-three X, O win rates at depths three, four, and five were 33.3%, 61.1%, 72.2%, respectively. With both sides at depth four and at depth five, O won 22.2%, 27.8%, respectively. The other four configurations used 2.4, 9.2, 3.8, 18.8 times the baseline mean node count, respectively. At equal depth five, draws accounted for 50.0% and O avoided defeat in 77.8% of games. O won less often than in the depth-three control but also lost less often; win rate alone does not establish deterioration.

Table 12: All extended-game openings; columns give (d_X, d_O), D means draw.

Start	First	$(3,3)$	$(3,4)$	$(3,5)$	$(4,4)$	$(5,5)$
0	O	O	O	O	X	D
0	X	D	O	O	D	O
1	O	D	O	O	X	O
1	X	D	O	D	X	X
2	O	O	O	O	O	D
2	X	X	D	X	X	D
3	O	D	D	X	D	D
3	X	D	X	O	X	D
4	O	X	O	O	O	O
4	X	O	X	X	O	X
5	O	X	O	D	D	D
5	X	O	O	O	X	D
6	O	O	D	O	D	D
6	X	X	O	O	X	O
7	O	X	O	O	O	O
7	X	D	X	O	D	D
8	O	X	O	O	D	X
8	X	O	X	O	X	X

Reference times use one sequential rerun per configuration of board zero with X first after all batch experiments finish. They are not eighteen-game means and depend on system load and caching. Node means cover eighteen games. Fixing X isolates changes to O's search configuration; equal-depth runs change both players' horizons. Interpret wins, losses, and draws together: additional draws can reduce win rate without adding losses. Results cover these fixed evaluations and eighteen deterministic openings, not arbitrary opposing strategies.

Statement of Contribution

Table 13: Group member contributions.

Member	Responsibility
Wang Libin	CSP formulation, the three tiling search policies, node-count comparisons, correctness checks, and Section I.
Jiang Jiahao	Designed evaluation and tuning, automatic and synthetic matchups, global-rule and depth experiments, Section III, and Extra Credit.
Wang Zizheng	Minimax and alpha–beta review, predefined matchups, rule and node-count verification, and Section II.
Xue Xiaokang	Human experiment and GUI workflow review, outcome and board analysis, Section IV, bilingual proofreading, and submission checks.